

PERSONAL STATEMENT

Starting from my fascination with the coordination of hardware when assembling a computer, to my understanding of the hierarchical structure of software systems, my technical understanding has always revolved around a core proposition. This concentrated on making complex systems operate reliably and efficiently and solving practical problems. This practice-driven way of thinking has made me no longer satisfied with merely achieving the predetermined functions. I began to wonder if can the system autonomously respond when the rules are unclear? And can it extract value from massive and disordered data? It was the exploration of these deeper questions that made me acutely aware that artificial intelligence is precisely the core engine to address these challenges. It endows the system with the ability to understand and make decisions, transforming passive execution tools into active problem-solving partners. Realizing this, I am even more eager to stand at the forefront of this transformation. My goal is to turn technology, from an entrepreneur's perspective, into products and services that truly improve people's lives, more than just participating in the design of intelligent systems as a developer; it also includes.

During my undergraduate studies, I systematically built a knowledge system in information technology and AI, and deepened my understanding through the integration of theory and practice. Courses such as Introduction to Software Engineering and Software System Design and Architecture have enabled me to master the overall process of the software development lifecycle, as well as understand the logic of system architecture design, modular management, and network communication. These courses taught me how to analyze problems from a system perspective, design efficient and scalable solutions, and maintain structured thinking in complex projects. In courses closely related to AI, such as *Data Structures and Algorithms*, and *Artificial Intelligence Foundation*, I gradually developed an algorithm-centered way of thinking and understood the entire process from data input and model training to result output. These courses strengthened my abilities in logical reasoning, computational complexity analysis, and algorithm optimization. They also allowed me to understand the working principles of AI systems from the underlying mechanisms.

Beyond the classroom, I continued to deepen my knowledge through project practice. In a traffic inquiry system developed with classmates, I was responsible for frontend development based on the Vue framework. Faced with a completely unfamiliar framework, I read official documents, analyzed open-source projects, and used AI tools for learning. I quickly mastered its component-based logic and data interaction mechanisms, and successfully completed the development task. This experience made me deeply appreciate the mutual promotion between theory and practice: theory provided me with ideas to solve problems, while practice allowed me to verify and optimize these ideas in real scenarios. To further consolidate my knowledge, I built my own GitHub repository, which systematically records the source code of projects I participated in, study notes, and development insights. This facilitated the creation of a sustainable learning system while enabling me to rapidly learn and innovate with new technologies. What's more, during a one-month server operation and maintenance internship, I gained a deep understanding of the integrity and complexity of computer systems. My main task involved developing an auto-update script for a bastion host (JumpServer) to facilitate its automatic "drift" to a new server upon failure. Initially, I planned to detect the new environment by checking for mismatched system IDs in configuration files. However, I discovered these files updated automatically during migration, rendering this method ineffective. After multiple attempts and debugging, my solution was to have the script, triggered at system startup, query the cloud provider's API using the machine's ID to obtain its purchase time. If this time was very recent (e.g., within minutes) compared to the script's execution time, it confirmed the environment shift, prompting the script to execute the necessary configuration updates. This process not only successfully ensured the migration but also made me realize that the various modules of a system are deeply interconnected, not isolated. At the same time, when writing backup scripts for the bastion host, I accidentally deleted configuration files due to an operational error. This mistake made me acutely aware of the critical importance of permission management and risk control in system design and operation. More importantly, whenever I encountered unsolvable errors and sought help from seniors, they would not give me the answer directly. Instead, they

Applicant: LUO Xiaoyu

Program: MSc Artificial Intelligence, City University of Hong Kong

guided me to track the root cause of the problem by recording logs and analyzing execution paths. This problem-oriented solution helped me gradually master methods for independent analysis, iterative improvement, and experience summarization.

These practical experiences have strengthened my determination to pursue further studies in the MSc Artificial Intelligence program at City University of Hong Kong. I learned that your program offers a solid theoretical foundation, while holding a strong application orientation, which perfectly aligns with my career plan. This program's curriculum comprehensively covers the core areas of AI such as machine learning, deep learning, natural language processing, and computer vision. Additionally, it also includes expansion courses like data mining. These will provide a clear path for me to build a systematic knowledge framework. Among them, the courses Artificial Intelligence and Intelligent Systems are the core of my study plan. I am eager to acquire the knowledge of knowledge representation, automatic reasoning, and planning algorithms through systematic learning, in order to make up for my current deficiencies in ability. To be specific, in previous projects, I mainly relied on a single machine learning model for prediction. However, these two courses will teach me how to enable multiple intelligent agents to complete more complex tasks through collaboration and competition. This will enable me to design and develop next-generation intelligent applications that truly possess autonomous decision-making and problem-solving capabilities in the future, rather than merely achieving the function of prediction.

Most importantly, this program is equipped with an industry-leading high-performance GPU computing cluster and a dedicated AI development platform. These advanced experimental facilities will not only support me in conducting large-scale deep learning and multi-modal data processing but also provide me with a complete practice environment from algorithm development to system deployment. With these, I will be able to cultivate my engineering capabilities for solving complex AI problems in a real technological ecosystem.

After systematic academic training and diverse practical experiences, I have a clear plan for my future career direction. After graduation, I plan to join an AI technology-driven enterprise, working on AI system development and innovative product design to accumulate comprehensive experience in technology R&D and project management. Within 3 to 5 years, I hope to establish an innovative startup company centered on AI technology, focusing on applying AI to the intelligent solution of real-world problems—such as urban traffic optimization, enterprise information security, and automated system construction.



華南師範大學
SOUTH CHINA NORMAL UNIVERSITY

Professor&Dean

Aberdeen Institute of Data Science and Artificial Intelligence, South China Normal University

Tel: +86 13760895528

Email: zengbiqing@scnu.edu.cn

Address: Taoyuan West Road, Nanhai Software Information Park, Shishan Town, Nanhai District, 528225, Foshan City

Recommendation Letter

Date: September 23, 2025

To whom It May Concern,

It is my pleasure to write this recommendation letter for LUO Xiaoyu, one of my best students in the course *Introduction to Computer Science and Technology*, to support him in applying for this esteemed postgraduate program at your university.

I came into acquaintance with Xiaoyu in September 2022 in my class, during which he fully showed his intelligence and concentration in class, behaving in a friendly and constructive manner in inner-group cooperation. In the process of interacting with him, both inside and outside the classroom, I have come into acquaintance with his solid theoretical foundation and problem-solving ability. He demonstrated his systematic understanding of the parts like computer system architecture, data representation, algorithm principles, programming languages, equipping himself with a fundamental cognitive framework of computer science. He also actively participated in diverse algorithm design and programming practices involving network programming, concurrent programming, and performance optimization, to cultivate his ability tailored to solve complex problems.

As I recalled, he took the initiative to participate in numerous programming practices, during which he wrote simple programs to analyze the machine-level representation of data and the process of instruction execution. He was able to use the GDB debugging tool to track the changes of program variables and understand the correspondence between high-level languages and machine instructions. Simultaneously, he was also adept at precisely reverse-analyzing the binary programs, and deeply studied the buffer overflow attacks and program linking mechanisms. Such things enhanced his understanding of the underlying principles of computer systems and refined his competencies in programming and system design, earning him a good score--86.

In a nutshell, it is without any reservation that I recommend this outstanding individual for admission to your esteemed program. I believe that he will continue to make positive contributions to this program, and the academic environment and lived experience in your university will be very valuable to his development.

If you need further information, please feel free to contact me.

Sincerely

Signature: